Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Nested Radical Denominator Integral

Evaluate the definite integral:

$$\int_2^6 \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}-1}\,dx$$

Solution: Let the infinite nested radical be denoted by the variable y:

$$y = \sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}$$

By squaring both sides, we observe the self-similar recursive structure of the expression:

$$y^2 = x + \sqrt{x+\sqrt{x+\cdots}} = x + y$$

Rearranging this algebraic relation gives x explicitly as a quadratic function of y:

$$x = y^2 - y$$

Differentiating both sides with respect to y yields the differential element dx:

$$dx = (2y-1)\,dy$$

We now determine the corresponding limits of integration in terms of y by solving $y^2 - y - x = 0$ (taking the positive root $y > 0$):

- **Lower limit ($x = 2$):**

$$y^2 - y - 2 = 0 \implies (y-2)(y+1) = 0 \implies y = 2$$

- **Upper limit ($x = 6$):**

$$y^2 - y - 6 = 0 \implies (y-3)(y+2) = 0 \implies y = 3$$

Substituting y, dx, and the new limits into the original definite integral, we observe that the denominator simplifies remarkably:

$$\int_2^6 \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}-1}\,dx = \int_2^3 \frac{1}{2y-1}(2y-1)\,dy$$

The factor $(2y-1)$ cancels completely from the numerator and denominator:

$$\int_2^3 1\,dy = [y]_2^3 = 3 - 2 = 1$$

The exact value of the integral is:

$$\int_2^6 \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}-1}\,dx = 1$$

1.2 Hard Integral

Floor Function Rational Integral

Evaluate the definite integral:

$$\int_1^{\infty} \frac{\lfloor x \rfloor}{x(x+1)^2}\, dx$$

Solution: Let the integral over the interval $[1, \infty)$ be denoted by I. Since the floor function $\lfloor x \rfloor$ takes the constant integer value n on each interval $[n, n+1)$ for $n \in \mathbb{Z}_{\geq 1}$, we partition the integration domain into unit intervals:

$$I = \sum_{n=1}^{\infty} \int_n^{n+1} \frac{\lfloor x \rfloor}{x(x+1)^2}\, dx = \sum_{n=1}^{\infty} n \int_n^{n+1} \frac{1}{x(x+1)^2}\, dx$$

We first find the antiderivative of the rational integrand via partial fraction decomposition:

$$\frac{1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

Clearing the denominator gives:

$$1 = A(x+1)^2 + Bx(x+1) + Cx$$

Setting specific values for x:

- For $x = 0$: $1 = A(1)^2 \implies A = 1$.
- For $x = -1$: $1 = C(-1) \implies C = -1$.
- Comparing the coefficient of x^2: $A + B = 0 \implies B = -1$.

Thus, the partial fraction expansion is:

$$\frac{1}{x(x+1)^2} = \frac{1}{x} - \frac{1}{x+1} - \frac{1}{(x+1)^2}$$

Integrating term by term yields the antiderivative $F(x)$:

$$F(x) = \int \frac{1}{x(x+1)^2}\, dx = \ln\left(\frac{x}{x+1}\right) + \frac{1}{x+1}$$

Evaluating the definite integral on the subinterval $[n, n+1]$:

$$\int_n^{n+1} \frac{1}{x(x+1)^2}\, dx = F(n+1) - F(n)$$

Substituting this back into the infinite series for I:

$$I = \sum_{n=1}^{\infty} n\,[F(n+1) - F(n)]$$

We consider the N-th partial sum and apply summation by parts (telescoping):

$$\sum_{n=1}^{N} n\left[F(n+1)-F(n)\right] = 1[F(2)-F(1)]+2[F(3)-F(2)]+\cdots+N[F(N+1)-F(N)]$$

$$= NF(N+1) - \sum_{n=1}^{N} F(n)$$

Let us analyze the asymptotic behavior of the boundary term $NF(N+1)$ as $N \to \infty$:

$$F(N+1) = \ln\left(1-\frac{1}{N+2}\right)+\frac{1}{N+2} \approx -\frac{1}{N+2}-\frac{1}{2(N+2)^2}+\frac{1}{N+2} = -\frac{1}{2(N+2)^2}$$

Therefore, $\lim_{N\to\infty} NF(N+1) = 0$. The integral is thus given by the negative sum of $F(n)$:

$$I = -\sum_{n=1}^{\infty} F(n) = \sum_{n=1}^{\infty}\left[\ln\left(\frac{n+1}{n}\right)-\frac{1}{n+1}\right]$$

We rewrite the series by adding and subtracting $\frac{1}{n}$ inside the summand:

$$I = \sum_{n=1}^{\infty}\left[\left(\frac{1}{n}-\frac{1}{n+1}\right)-\left(\frac{1}{n}-\ln\left(1+\frac{1}{n}\right)\right)\right]$$

Splitting this into two separate convergent series:

$$I = \sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)-\sum_{n=1}^{\infty}\left[\frac{1}{n}-\ln\left(1+\frac{1}{n}\right)\right]$$

The first series is a standard telescoping sum evaluating to 1:

$$\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right) = \left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots = 1$$

The second series is precisely the classic series definition of the Euler-Mascheroni constant $\gamma \approx 0.5772156649$:

$$\gamma = \sum_{n=1}^{\infty}\left[\frac{1}{n}-\ln\left(1+\frac{1}{n}\right)\right]$$

Combining both parts, the exact closed-form value of the integral is:

$$I = 1-\gamma$$

Differentials

2.1 Medium Differential

Non-Linear Second-Order Differential Equation

Let $f(x)$ be a twice-differentiable function satisfying:

$$f''(x) + f(x)[f'(x)]^3 = [f'(x)]^2$$

Given the initial conditions $f(1) = 0$ and $f'(1) = 1$, compute the exact value of $f\left(\frac{5}{2} - \frac{2}{e}\right)$.

Solution: We solve this non-linear differential equation by treating the dependent variable $y = f(x)$ as the independent variable, and $x = x(y)$ as the function to be determined. By standard inverse function differentiation rules:

$$f'(x) = \frac{dy}{dx} = \frac{1}{x'(y)}$$

Differentiating this relation with respect to x using the chain rule:

$$f''(x) = \frac{d}{dx}\left(\frac{1}{x'(y)}\right) = \frac{dy}{dx}\frac{d}{dy}\left(\frac{1}{x'(y)}\right) = \frac{1}{x'(y)}\left(-\frac{x''(y)}{[x'(y)]^2}\right) = -\frac{x''(y)}{[x'(y)]^3}$$

We now substitute these inverse derivative identities into the original differential equation:

$$-\frac{x''(y)}{[x'(y)]^3} + y\left(\frac{1}{x'(y)}\right)^3 = \left(\frac{1}{x'(y)}\right)^2$$

Multiplying both sides through by $[x'(y)]^3$ (since $f'(x) \neq 0$):

$$-x''(y) + y = x'(y) \implies x''(y) + x'(y) = y$$

This transforms the highly non-linear equation into a straightforward second-order linear ordinary differential equation with constant coefficients for $x(y)$. We convert the initial conditions at $x = 1$:

$$f(1) = 0 \implies x(0) = 1$$

$$f'(1) = 1 \implies x'(0) = \frac{1}{f'(1)} = 1$$

We now solve the linear ODE $x''(y) + x'(y) = y$. The characteristic equation for the homogeneous part $x''(y) + x'(y) = 0$ is $r^2 + r = 0$, giving roots $r = 0$ and $r = -1$. Therefore, the complementary solution is:

$$x_c(y) = C_1 + C_2 e^{-y}$$

For the particular solution $x_p(y)$, we try a polynomial of degree 2, $x_p(y) = Ay^2 + By$:

$$x_p'(y) = 2Ay + B, \qquad x_p''(y) = 2A$$

Substituting into the ODE:

$$2A + 2Ay + B = y \implies 2Ay + (2A + B) = y$$

Equating coefficients of like powers of y:

$$2A = 1 \implies A = \frac{1}{2}$$

$$2A + B = 0 \implies 1 + B = 0 \implies B = -1$$

Thus, the general solution for $x(y)$ is:

$$x(y) = C_1 + C_2 e^{-y} + \frac{1}{2}y^2 - y$$

We apply the initial conditions $x(0) = 1$ and $x'(0) = 1$ to determine the constants C_1 and C_2:

$$x(0) = C_1 + C_2 = 1$$

$$x'(y) = -C_2 e^{-y} + y - 1 \implies x'(0) = -C_2 - 1 = 1 \implies C_2 = -2$$

Using $C_1 + C_2 = 1$:

$$C_1 - 2 = 1 \implies C_1 = 3$$

The explicit relationship for the inverse function $x(y)$ is therefore:

$$x(y) = 3 - 2e^{-y} + \frac{1}{2}y^2 - y$$

To find $f\left(\frac{5}{2} - \frac{2}{e}\right)$, we must find the unique value y_0 such that $x(y_0) = \frac{5}{2} - \frac{2}{e}$. We test $y = 1$:

$$x(1) = 3 - 2e^{-1} + \frac{1}{2}(1)^2 - 1 = 2 + \frac{1}{2} - \frac{2}{e} = \frac{5}{2} - \frac{2}{e}$$

Since $x'(y) = 2e^{-y} + y - 1 > 0$ for all $y \geq 0$, the function $x(y)$ is strictly increasing, guaranteeing that $y = 1$ is the unique solution. Hence:

$$f\left(\frac{5}{2} - \frac{2}{e}\right) = 1$$

2.2 Hard Differential

Rational Homogeneous Differential Equation

Solve the differential equation:

$$\frac{dy}{dx} = \frac{x + y - 4}{x - y - 6}$$

Given that the curve passes through the point $(6, -1)$, find the other integral x-value where the curve crosses $y = -1$.

Solution: We first determine the intersection point (h, k) of the two linear expressions in the numerator and denominator by solving the simultaneous system:

$$x + y - 4 = 0$$

$$x - y - 6 = 0$$

Adding the two equations yields $2x - 10 = 0 \implies x = 5$. Subtracting the second from the first gives $2y + 2 = 0 \implies y = -1$. We shift the coordinate system to this intersection point by defining translated variables u and v:

$$u = x - 5 \implies dx = du$$

$$v = y + 1 \implies dy = dv$$

In terms of u and v, the differential equation simplifies into a standard homogeneous equation:

$$\frac{dv}{du} = \frac{u + v}{u - v}$$

We solve this by introducing the substitution $v = tu$, where $t = \frac{v}{u}$ and $\frac{dv}{du} = t + u\frac{dt}{du}$:

$$t + u\frac{dt}{du} = \frac{u + tu}{u - tu} = \frac{1 + t}{1 - t}$$

Isolating the derivative term $u\frac{dt}{du}$:

$$u\frac{dt}{du} = \frac{1 + t}{1 - t} - t = \frac{1 + t - t(1 - t)}{1 - t} = \frac{1 + t^2}{1 - t}$$

Separating the variables u and t:

$$\frac{1 - t}{1 + t^2}\,dt = \frac{du}{u}$$

We integrate both sides by splitting the fraction on the left:

$$\int \left(\frac{1}{1 + t^2} - \frac{t}{1 + t^2} \right) dt = \int \frac{du}{u}$$

$$\tan^{-1}(t) - \frac{1}{2}\ln(1 + t^2) = \ln|u| + C$$

We now apply the given initial condition to determine the constant of integration C. The curve passes through $(x, y) = (6, -1)$. In our translated coordinates:

$$u_0 = 6 - 5 = 1, \qquad v_0 = -1 + 1 = 0 \implies t_0 = \frac{v_0}{u_0} = \frac{0}{1} = 0$$

Substituting $t = 0$ and $u = 1$ into our integrated solution:

$$\tan^{-1}(0) - \frac{1}{2}\ln(1 + 0^2) = \ln|1| + C \implies 0 - 0 = 0 + C \implies C = 0$$

Thus, the specific implicit equation governing the integral curve is:

$$\tan^{-1}\left(\frac{v}{u}\right) - \frac{1}{2}\ln\left(1 + \frac{v^2}{u^2}\right) = \ln|u|$$

We can simplify the logarithmic terms using logarithmic identities:

$$\tan^{-1}\left(\frac{v}{u}\right) = \ln|u| + \frac{1}{2}\ln\left(\frac{u^2+v^2}{u^2}\right) = \ln|u| + \ln\left(\sqrt{u^2+v^2}\right) - \ln|u| = \ln\left(\sqrt{u^2+v^2}\right)$$

In terms of polar coordinates centered at $(5, -1)$ where $u = r\cos\theta$ and $v = r\sin\theta$, this is simply the logarithmic spiral $r = e^{\theta}$. We are tasked with finding the other integral x-value where the curve crosses the horizontal line $y = -1$. When $y = -1$, the vertical translated variable vanishes: $v = 0$. Substituting $v = 0$ (and thus $t = 0$) back into our implicit algebraic equation:

$$\tan^{-1}(0) - \frac{1}{2}\ln(1+0) = \ln|u| \implies \ln|u| = 0 \implies |u| = 1$$

This absolute value equation yields precisely two roots for u:

- $u = 1 \implies x = u + 5 = 6$ (which corresponds to the given initial point $(6, -1)$).
- $u = -1 \implies x = u + 5 = 4$.

Therefore, the other integer x-value where the curve crosses $y = -1$ is:

$$x = 4$$

Derivatives

3.1 Medium Derivative

Implicit Function Limit Evaluation

If $(x+f(x))^2 = xf(x)+3$, compute:

$$\lim_{x\to 2} f'(x)(x+2f(x)).$$

Solution: We begin by expanding the algebraic relation defined by the problem statement:

$$x^2 + 2xf(x) + [f(x)]^2 = xf(x) + 3$$

Combining like terms by subtracting $xf(x)$ from both sides:

$$[f(x)]^2 + xf(x) + x^2 - 3 = 0$$

We differentiate both sides of this equation implicitly with respect to x:

$$\frac{d}{dx}\left([f(x)]^2 + xf(x) + x^2 - 3\right) = 0$$

$$2f(x)f'(x) + \left(f(x) + xf'(x)\right) + 2x = 0$$

We factor out $f'(x)$ from the derivative terms:

$$f'(x)\,(2f(x)+x) + f(x) + 2x = 0$$

Rearranging the equation to isolate the target product $f'(x)(x+2f(x))$:

$$f'(x)(x+2f(x)) = -(2x+f(x))$$

This identity holds for all x in the domain of differentiability of f. Therefore, evaluating the required limit as $x \to 2$ simply reduces to evaluating the continuous right-hand side:

$$\lim_{x\to 2} f'(x)(x+2f(x)) = \lim_{x\to 2} -(2x+f(x)) = -(4+f(2))$$

To complete the calculation, we determine the value of $f(2)$ by substituting $x = 2$ directly into the simplified original equation $[f(x)]^2 + xf(x) + x^2 - 3 = 0$:

$$[f(2)]^2 + 2f(2) + 2^2 - 3 = 0$$

$$[f(2)]^2 + 2f(2) + 1 = 0 \implies (f(2)+1)^2 = 0 \implies f(2) = -1$$

Substituting $f(2) = -1$ into our evaluated limit expression:

$$\lim_{x\to 2} f'(x)(x+2f(x)) = -(4+(-1)) = -3$$

The exact value of the limit is:

$$-3$$

3.2 Hard Derivative

High-Order Derivative Recurrence Relation

A smooth real function $f(x)$ satisfies

$$f(x) + f'(x) + f''(x) = 2026 \quad \text{for all } x, \text{ and}$$

has a local minimum at $(26, 20)$. Find $f^{(20)}(26)$.

Solution: Let us analyze the information given at the point $x = 26$. Because $f(x)$ has a local minimum at $(26, 20)$, we possess two fundamental boundary conditions at this point:

$$f(26) = 20$$
$$f'(26) = 0$$

We evaluate the given second-order differential equation at $x = 26$:

$$f''(26) + f'(26) + f(26) = 2026$$

Substituting our initial values $f(26) = 20$ and $f'(26) = 0$:

$$f''(26) + 0 + 20 = 2026 \implies f''(26) = 2006$$

To determine higher-order derivatives, we differentiate the differential equation $f''(x) + f'(x) + f(x) = 2026$ successively with respect to x. For any integer $n \geq 1$, differentiating n times causes the constant right-hand side (2026) to vanish:

$$f^{(n+2)}(x) + f^{(n+1)}(x) + f^{(n)}(x) = 0 \quad \text{for all } n \geq 1$$

Let us define the numerical sequence $a_n = f^{(n)}(26)$ representing the n-th derivative evaluated at $x = 26$. The recurrence relation for $n \geq 1$ is:

$$a_{n+2} = -a_{n+1} - a_n$$

We compute the first several terms of the sequence (a_n) starting from $n = 1$:

- $a_1 = f'(26) = 0$
- $a_2 = f''(26) = 2006$
- $a_3 = -a_2 - a_1 = -2006 - 0 = -2006$
- $a_4 = -a_3 - a_2 = -(-2006) - 2006 = 0$
- $a_5 = -a_4 - a_3 = -0 - (-2006) = 2006$
- $a_6 = -a_5 - a_4 = -2006 - 0 = -2006$
- $a_7 = -a_6 - a_5 = -(-2006) - 2006 = 0$

We observe that for all $n \geq 1$, the derivative sequence is strictly periodic with period 3:

$$a_{3k+1} = 0, \qquad a_{3k+2} = 2006, \qquad a_{3k} = -2006 \qquad \text{for } k \in \mathbb{Z}_{\geq 0}$$

We need to evaluate the 20th derivative, which corresponds to a_{20}. We reduce the index modulo 3:

$$20 = 3(6) + 2 \implies 20 \equiv 2 \pmod 3$$

Therefore, a_{20} corresponds to the second element in the periodic cycle:

$$f^{(20)}(26) = a_{20} = a_2 = 2006$$

The exact value is:

$$f^{(20)}(26) = 2006$$

Physics & Engineering

4.1 Mechanics

Hammerhead Shark Dorsal Lift Optimization

Great hammerhead sharks regularly swim tilted at an angle θ to take advantage of lift generated by their large dorsal fin. The dorsal fin experiences 3 times more normal force than the pectoral fins, which are both angled 75° from the vertical. Calculate the optimal angle θ (in degrees) that produces the most lift. Consider the dorsal and pectoral fins only.

Solution: When a shark is in its standard upright swimming orientation ($\theta = 0^\circ$), its dorsal fin lies entirely in the vertical sagittal plane. Consequently, the normal force generated by the dorsal fin points horizontally to the side, contributing zero vertical lift. When the shark rolls around its longitudinal axis by a tilt angle θ, the normal vector of the dorsal fin is rotated by θ from the horizontal plane (or $90^\circ - \theta$ from the vertical). Therefore, the vertical lift force L_D contributed by the dorsal fin is:

$$L_D(\theta) = N_D \cos(90^\circ - \theta) = N_D \sin\theta$$

where N_D is the magnitude of the normal force on the dorsal fin.

We now analyze the lift contributed by the two pectoral fins. When upright ($\theta = 0^\circ$), the two pectoral fins extend symmetrically outward from the body, with their normal vectors angled at $\beta = 75^\circ$ from the vertical. As the shark rolls by angle θ in the coronal plane, one pectoral fin rotates toward the vertical while the other rotates toward the horizontal. Their new angles from the vertical become $(75^\circ - \theta)$ and $(75^\circ + \theta)$. Let N_{pec} denote the normal force magnitude on each individual pectoral fin. The total vertical lift L_P generated by both pectoral fins as a function of roll angle θ is:

$$L_P(\theta) = N_{\text{pec}} \cos(75^\circ - \theta) + N_{\text{pec}} \cos(75^\circ + \theta)$$

Using the trigonometric sum-to-product identity $\cos(A - B) + \cos(A + B) = 2\cos A \cos B$:

$$L_P(\theta) = 2N_{\text{pec}} \cos(75^\circ) \cos\theta$$

Combining the lift from all three fins, the total vertical lift $L(\theta)$ is given by:

$$L(\theta) = N_D \sin\theta + 2N_{\text{pec}} \cos(75^\circ) \cos\theta$$

We are given that "the dorsal fin experiences 3 times more normal force than the pectoral fins". Interpreting this as the ratio of the dorsal fin force to each individual pectoral fin ($N_D = 3N_{\text{pec}}$), we substitute $N_D = 3N_{\text{pec}}$ into the lift equation:

$$L(\theta) = N_{\text{pec}} [3\sin\theta + 2\cos(75^\circ) \cos\theta]$$

To find the roll angle θ that maximizes this lift function of the general form $A\sin\theta + B\cos\theta$, we differentiate with respect to θ and set the derivative to zero:

$$\frac{dL}{d\theta} = N_{\text{pec}} [3\cos\theta - 2\cos(75^\circ) \sin\theta] = 0$$

$$3\cos\theta = 2\cos(75^\circ)\sin\theta \implies \tan\theta = \frac{3}{2\cos(75^\circ)}$$

We evaluate the exact algebraic and numerical value of this trigonometric ratio using $\cos(75^\circ) = \frac{\sqrt{6}-\sqrt{2}}{4} \approx 0.258819$:

$$2\cos(75^\circ) = \frac{\sqrt{6}-\sqrt{2}}{2} \approx 0.517638$$

$$\tan\theta = \frac{3}{0.517638} \approx 5.79555$$

Taking the inverse tangent yields the optimal roll angle:

$$\theta = \tan^{-1}(5.79555) \approx 80.21^\circ$$

Note: If the problem statement's comparison is instead interpreted as the dorsal fin force being 3 times the *combined* normal force of both pectoral fins together ($N_D = 3N_{\text{total}} = 6N_{\text{pec}}$), the optimality condition becomes $\tan\theta = \frac{3}{\cos(75^\circ)} \approx 11.591$, which yields $\theta \approx 85.07^\circ$. Under the primary standard interpretation per pectoral fin, the optimal angle is:

$$\theta \approx 80.2^\circ$$

4.2 Quantum Mechanics

Infinite Potential Well Energy Transition

An electron is confined in a one-dimensional infinite potential well. The energy difference between the $n = 3$ and $n = 2$ states is 7.20 eV. Calculate the width of the well in units of 10^{-10} m. Use $h = 6.626 \times 10^{-34}$ J s, $m_e = 9.109 \times 10^{-31}$ kg, and 1 eV $= 1.602 \times 10^{-19}$ J.

Solution: The quantized energy levels E_n of a particle of mass m_e confined within a one-dimensional infinite potential well of width L are given by the standard formula:

$$E_n = \frac{n^2h^2}{8m_eL^2}$$

The energy difference ΔE between the $n = 3$ and $n = 2$ quantum states is:

$$\Delta E = E_3 - E_2 = \frac{(3^2 - 2^2)h^2}{8m_eL^2} = \frac{5h^2}{8m_eL^2}$$

We rearrange this expression to solve explicitly for the well width L:

$$L = \sqrt{\frac{5h^2}{8m_e\Delta E}}$$

We first convert the given transition energy $\Delta E = 7.20$ eV into SI units of Joules:

$$\Delta E = (7.20\text{ eV}) \times \left(1.602 \times 10^{-19}\text{ J/eV}\right) = 1.15344 \times 10^{-18}\text{ J}$$

We now substitute the physical constants into the numerator and denominator:

- **Numerator ($5h^2$):**

$$h^2 = \left(6.626 \times 10^{-34}\,\text{J s}\right)^2 = 43.903876 \times 10^{-68}\,\text{J}^2\,\text{s}^2$$

$$5h^2 = 5 \times \left(43.903876 \times 10^{-68}\right) = 2.1951938 \times 10^{-65}\,\text{J}^2\,\text{s}^2$$

- **Denominator ($8m_e\Delta E$):**

$$8m_e = 8 \times \left(9.109 \times 10^{-31}\,\text{kg}\right) = 7.2872 \times 10^{-30}\,\text{kg}$$

$$8m_e\Delta E = \left(7.2872 \times 10^{-30}\,\text{kg}\right) \times \left(1.15344 \times 10^{-18}\,\text{J}\right) = 8.405347968 \times 10^{-48}\,\text{J kg}$$

Dividing the numerator by the denominator inside the square root:

$$L^2 = \frac{2.1951938 \times 10^{-65}}{8.405347968 \times 10^{-48}} \approx 2.611663 \times 10^{-19}\,\text{m}^2 = 26.11663 \times 10^{-20}\,\text{m}^2$$

Taking the principal square root yields:

$$L = \sqrt{26.11663 \times 10^{-20}\,\text{m}^2} \approx 5.11044 \times 10^{-10}\,\text{m}$$

Expressed in units of 10^{-10} m (Ångströms) to three significant figures, the width of the potential well is:

$$L \approx 5.11 \times 10^{-10}\,\text{m}$$

4.3 Electromagnetism

Grounded Conducting Plane Disk Induced Charge

A point charge $q > 0$ is located at a distance d along the z-axis at $(0, 0, d)$ above an infinite grounded conducting plane at $z = 0$. The induced surface charge density $\sigma(r)$ on the plane is given by:

$$\sigma(r) = -\frac{qd}{2\pi(r^2 + d^2)^{3/2}}$$

Evaluate the total induced charge Q_{ind} on the circular disk region defined by $0 \le r \le d\sqrt{3}$ and find the numerical value of the ratio:

$$I = -\frac{2Q_{\text{ind}}}{q}$$

Solution: To determine the total induced charge Q_{ind} accumulated within the circular disk of radius $R = d\sqrt{3}$ centered directly beneath the point charge at the origin, we integrate the surface charge density $\sigma(r)$ over the area of the disk using polar coordinates (r, θ):

$$Q_{\text{ind}} = \int_0^{2\pi} d\theta \int_0^{d\sqrt{3}} \sigma(r) r\,dr = 2\pi \int_0^{d\sqrt{3}} \left[-\frac{qd}{2\pi(r^2 + d^2)^{3/2}}\right] r\,dr$$

The 2π angular integration factor cancels with the 2π in the denominator of $\sigma(r)$:

$$Q_{\text{ind}} = -qd \int_0^{d\sqrt{3}} \frac{r}{(r^2 + d^2)^{3/2}}\, dr$$

We evaluate this integral using the substitution $u = r^2 + d^2$, which gives $du = 2r\, dr \implies r\, dr = \frac{1}{2} du$. We transform the integration boundaries accordingly:

- When $r = 0$: $u = 0^2 + d^2 = d^2$.
- When $r = d\sqrt{3}$: $u = (d\sqrt{3})^2 + d^2 = 3d^2 + d^2 = 4d^2$.

Substituting these into the definite integral:

$$Q_{\text{ind}} = -qd \int_{d^2}^{4d^2} u^{-3/2} \left(\frac{1}{2}\right) du = -\frac{qd}{2} \left[-2u^{-1/2}\right]_{d^2}^{4d^2}$$

The factor of $-\frac{1}{2}$ cancels with -2:

$$Q_{\text{ind}} = qd \left[\frac{1}{\sqrt{u}}\right]_{d^2}^{4d^2} = qd \left(\frac{1}{\sqrt{4d^2}} - \frac{1}{\sqrt{d^2}}\right) = qd \left(\frac{1}{2d} - \frac{1}{d}\right)$$

Combining the fractions inside the parentheses:

$$Q_{\text{ind}} = qd \left(-\frac{1}{2d}\right) = -\frac{q}{2}$$

We now compute the dimensionless numerical ratio I defined by the problem:

$$I = -\frac{2Q_{\text{ind}}}{q} = -\frac{2\left(-\frac{q}{2}\right)}{q} = -\frac{-q}{q} = 1$$

The exact numerical value of the ratio is:

$$I = 1$$